

MAT 535 HW10

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Problem 1 (§18.1#7). Let V be a 4-dimensional permutation module for S_4 described in Example 3 of the second set of examples. Let $\pi: D_8 \rightarrow S_4$ be the permutation representation of D_8 obtained from the action of D_8 by left multiplication on the set of left cosets of its subgroup $\langle s \rangle$. Make V into an FD_8 -module via π as described in Example 4 and write out the 4×4 matrices for r and s given by this representation with respect to the basis e_1, e_2, e_3, e_4 .

Solution. The left cosets of $\langle s \rangle$ are the subgroup itself, $\{r, rs\}$, $\{r^2, r^2s\}$, and $\{r^3, r^3s\}$, which we will say correspond respectively to the basis elements e_1, e_2, e_3, e_4 . Then, r acts by cyclically permuting the cosets, so the matrix corresponding to this action is

$$\rho(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Left multiplication by s preserves $\langle s \rangle$ and $\{r^2, r^2s\}$ and swaps $\{r, rs\}$ and $\{r^3, r^3s\}$ due to the fact $sr^k = r^{-k}s$, so the matrix describing the action of s is

$$\rho(s) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad \square$$

Problem 2 (§18.1#13). Let R be a ring and let M and N be simple R -modules.

- (a) Prove that every nonzero R -module homomorphism from M to N is an isomorphism.
- (b) Prove Schur's lemma: if M is a simple R -module then $\text{Hom}_R(M, M)$ is a division ring.

Solution. (a) The kernel of a nonzero homomorphism from M to N cannot be M , and the only other choice of submodule is 0, so the homomorphism is injective. Similarly, the image of the nonzero homomorphism cannot be zero, so it must be N , so the homomorphism is surjective.

(b) Every nonzero element of $\text{Hom}_R(M, M)$ is an isomorphism by (a), hence is invertible. $\square$

Problem 3 (§18.2#11). Prove that if R is a ring with 1 such that every R -module is free then R is a division ring.

Solution. We wish to show that each $a \in R - \{0\}$ has a two-sided inverse. Let $I = Ra$ be the left ideal generated by a ; by assumption, the R -modules I and R/I are both free. We can construct a section $R/I \rightarrow R$ of the projection $R \rightarrow R/I$ by choosing the images of a basis of R/I , so that the short exact sequence $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$ splits, where the map $I \rightarrow R$ is the inclusion. The ranks of I and R/I as R -modules thus add to the rank of R , which is 1; since $I \neq 0$ (it contains a), $R/I = 0$, so that the inclusion $I \rightarrow R$ is an isomorphism, which implies $I = R$. Thus multiplication on the right by a is a permutation of $R - \{0\}$ sending an element a^{-1} to 1; then right multiplication by a^{-1} sends a to 1 as well, so a^{-1} is the (two-sided) inverse of a . $\square$

Problem 4 (§18.2#12). Let F be a field, let $f(x) \in F[x]$, and let $R = F[x]/(f(x))$. Find necessary and sufficient conditions on the factorization of $f(x)$ in $F[x]$ so that R is a semisimple ring.

Solution. A commutative ring is semisimple if and only if it is a product of fields because a matrix ring over a division algebra is commutative if and only if it is a field. Let $f(x) = p_1(x)^{e_1} \cdots p_r(x)^{e_r}$ for pairwise distinct irreducible polynomials $p_i(x)$ and positive integers e_i . By the Chinese remainder theorem,

$$R \cong \prod_{i=1}^r F[x]/(p_i(x)^{e_i}).$$

A direct product of fields has no nonzero nilpotent elements, so each e_i must be 1 (else $p_1(x) \cdots p_r(x)$ would be a nonzero nilpotent), and in this case the factors given by the Chinese remainder theorem are indeed fields, so $f(x)$ being separable is necessary and sufficient for R to decompose as a product of fields. $\square$

Problem 5 (§18.3#13). Let G be the cyclic group of order n and let $R = \mathbb{Q}G$. Describe the Wedderburn decomposition of R and find the number and degrees of the irreducible representations of G over $\mathbb{Q}$. In particular, show that if $n = p$ is a prime then G has exactly one nontrivial irreducible representation over $\mathbb{Q}$ and that this representation has degree $p - 1$.

Solution. The group ring $\mathbb{Q}G$ is isomorphic to $\mathbb{Q}[x]/(x^n - 1)$. By the previous problem this ring is semisimple (since $x^n - 1$ is separable), and the factorization of $x^n - 1$ over $\mathbb{Q}$ is into cyclotomic polynomials, so

$$\mathbb{Q}G \cong \prod_{d|n} \mathbb{Q}[x]/(\Phi_d(x)).$$

Thus there is one irreducible representation of degree $\varphi(d)$ for each divisor d of n . In particular, when $n = p$ is prime, there is an irreducible representation of degree $\varphi(1) = 1$ and one of degree $\varphi(p) = p - 1$. $\square$